



Randomized Control Trials

Effect of artificial intelligence-based video-game system on dysphagia in patients with stroke: A randomized controlled trial

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SUMMARY

Background and aims: Post-stroke dysphagia is highly prevalent and causes complication. While video games have demonstrated potential to increase patient engagement in rehabilitation, their efficacy in stroke patients with dysphagia remains unclear. This aim of this study was to explore the effectiveness of the artificial intelligence-based video-game (AI-VG) intervention in improving swallowing function among stroke patients with dysphagia.

Methods: A prospective, single blinded, randomized controlled trial (RCT) was conducted in a rehabilitation hospital from October 2023 to July 2024. Participants were randomly assigned to either the AI-VG system group or the usual care group. All participants received training for 30 min per session per day, five times per week for 4 weeks. The primary outcome was change in swallowing function from baseline (T0) to post-intervention (T1) and 1 month follow-up (T2). Secondary outcomes included changes in laryngeal function, oral intake function, nutritional status, and swallowing-related quality of life. The adherence, satisfaction, and acceptance of the two groups were evaluated. Changes in outcomes over time were compared using generalized estimating equation modeling.

Results: A total of 84 participants (AI-VG group = 42, age = 65.00 ± 9.70 years, 57.10 % male; usual care group = 42, age = 66.40 ± 13.10 years, 69.00 % male) were included in this study. Compared with the usual care group, the AI-VG group showed significantly improved swallowing function, with a mean group difference of 4.02 (95 % CI = from −6.16 to −1.89, $P < 0.001$) at T1 and 4.14 (95 % CI = from −6.16 to 2.12, $P < 0.001$) at T2. Oral intake function, nutritional status, and swallowing-related quality of life improved significantly ($P < 0.001$ for overall group × time interaction). Adherence was significantly higher in the intervention group than in the control group (median [IQR], 18.00 [17.00–20.00] vs. 16.00 [15.00–17.00], $P < 0.001$). The intervention group had higher levels of acceptance and satisfaction of AI-VG (103.00 [100.50–104.00] and 73.00 [72.00–74.00], respectively). No significant difference was observed in laryngeal function ($P > 0.05$).

Conclusions: The AI-VG intervention is an effective method for enhancing swallowing function in patients with post-stroke dysphagia. Future validation through multicenter, large-sample RCTs are warranted to confirm the findings and explore broader clinical applications.

Clinical trial registration: ClinicalTrials. Gov Identifier: NCT05978700, <https://clinicaltrials.gov/study/NCT05978700>.

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1. Introduction

Dysphagia, a common post-stroke complication, affects up to 9.75 million people globally, with a prevalence ranging from 28 % to 75 % [1–3]. This condition leads to serious consequences, including aspiration pneumonia, malnutrition, long-term dependence, and choking [4,5]. Post-stroke dysphagia is the second leading risk factor for in-hospital mortality among patients with stroke [6], and it affects quality of life, increases the risk of depression, prolongs hospitalization, and increases healthcare costs [7,8].

Swallowing is a complex neuromuscular involving the coordination of multiple muscle groups to minimize residue while delivering food or liquid from the mouth to the stomach [9]. Traditional dysphagia rehabilitation methods, such as chin tuck against resistance (CTAR) exercise, respiratory muscle training, and oropharyngeal muscle training, have been widely used [10–12]. However, these approaches often face limitations, including monotonous training modes, transportation barriers, and a shortage of skilled professionals. These factors can result in low patient compliance, poor rehabilitation outcomes, and increased workload for healthcare professionals [9,13,14]. These interventions may not meet the needs of patients in an era of rapid technological advancement.

Advanced interactive technologies, such as artificial intelligence-based video games (AI-VG), offer promising new directions for the rehabilitation of patients with post-stroke dysphagia. The use of video games in rehabilitation through visualization can increase patient motivation, improve patient rehabilitation endurance, and enable transparent measurement of objective results of patient training [15]. Video games that combine AI can be more adaptive to participant behavior, accurately capture participant movements, assist in interactive character control, and enhance the overall gaming experience [16].

Several studies have explored the potential of game-based approaches in dysphagia rehabilitation. Park et al. [17] combined CTAR with video games and found remarkable higher enjoyment scores than in conventional therapy. Hou et al. [18] integrated gaming with surface electromyography biofeedback, demonstrating significantly improved swallowing function compared with usual care. However, these studies have some limitations such as the lack of patient usability evaluation or a narrow focus on single training maneuvers that may not meet the intensity and needs of dysphagia training. Additionally, these studies required additional equipment, such as triode electrodes or electronic resistance bar, thus potentially burdening patients. The present study developed an AI-VG system specifically designed to support stroke dysphagia training to address these limitations.

This randomized controlled trial (RCT) primarily aimed to determine the effect of the AI-VG system on swallowing function in patients with post-stroke dysphagia compared with usual care. The secondary aims were to assess the effect on laryngeal function, oral intake function, nutritional status, and swallowing-related quality of life. Furthermore, this study aimed to evaluate patient acceptability and satisfaction with the AI-VG system.

2. Materials and methods

2.1. Study design

This study was a prospective, single-blind RCT conducted from October 2023 to July 2024. This study followed the Consolidated Standards of Reporting Trials (CONSORT) reporting guidelines [19]. It was approved by the Institutional Review Board of The Hong

Kong Polytechnic University (Approval No. HSEARS20230502007) and the Ethics Review Committee of Beijing Xiaotangshan Hospital (Approval No. LS20230720-1). It was registered with [ClinicalTrials.gov](https://www.clinicaltrials.gov) (NCT05978700). A written informed consent was obtained from all participants before data collection to ensure confidentiality and anonymity.

2.2. Participants and setting

This study was conducted in a rehabilitation hospital in Beijing. The inclusion criteria were as follows: (1) diagnosis of stroke confirmed by clinical signs and neuroimaging evidence, clinical diagnosis based on the American Heart Association/American Stroke Association statement, and neuroimaging confirmation based on computed tomography (CT) or magnetic resonance imaging (MRI) images [20]; (2) diagnosed as dysphagia by the Gugging Swallowing Screen (GUSS); (3) first cerebral hemispheric onset and the course of the disease is less than 1 year; (4) age over than 18 years; (5) no significant cognitive impairment, with Mini-Mental State Examination (MMSE) score ≥ 24 and able to execute instructions correctly; and (6) provide informed written consent. The exclusion criteria included (1) mental disease or cognitive impairment, (2) warning signs of possible cancer or serious disease requiring emergency treatment, (3) epilepsy or visual impairment preventing computer use, or (4) pregnancy or lactation.

2.3. Sample size

The sample size was estimated using G-Power 3.1 [21], based on the pilot study results using GUSS, with means of 16.79 and 12.83 and standard deviations (SDs) of 4.73 and 6.65 for the intervention and control groups, respectively. The effect size (Cohen's *d*) of the pilot study was 0.75. A total of 32 participants were estimated in each group for a significant level of 0.05 and a power of 85 %. The sample size for each group was increased by 20 % to avoid loss of power due to potential dropout, resulting in 78 participants.

2.4. Randomization and blinding

A research assistant not involved in the study performed randomization using R software, and a sealed opaque envelope was used to preserve the order of the randomized groups [22]. The participants were assigned to the AI-VG group or usual care group on the basis of the number in the envelope after inclusion.

The assessor was blinded to participant allocation and not involved in the treatment nor statistical analyses. Blinding participants and interventionists was not possible due to the nature of the AI-VG intervention. The participants were instructed not to disclose their groups nor the intervention details to assessor or others.

2.5. Intervention description

All groups' interventions were conducted for 30 min a day, five times a week, for a total of 4 weeks. The experimental group received the AI-VG intervention, and the control group received the usual care treatment. Both groups involved strict monitoring of vital signs, maintaining a supportive ward environment, routine airway humidification, and nutritional management.

2.5.1. AI-VG group

The AI-VG group received dysphagia rehabilitation training using the AI-VG system. Each participant was provided with a

computer equipped with the AI-VG system and seated facing a screen with an overhead camera. The camera recorded real-time facial changes during the training process. The AI-VG system utilized a facial landmark detector from the MediaPipe AI model library, capable of generating 3D positions of 468 facial feature points on facial images [23]. The system detected changes in the participant's facial muscles by recognizing changes in the 3D position of the facial feature points, providing biofeedback through real-time game control computation.

The AI-VG system comprised three games targeting specific exercises, as shown in Fig. 1: (1) “Collecting Carrots”: Focused on lip exercises. Participants control the in-game character by puffing out their cheeks, alternating between the left and right sides, holding each puff for 2 or 3 s. As the game character moves left and right to reach the target, carrots automatically drop into its backpack. Participants must complete 15 repetitions of puffing each cheek. (2) “Out of the Maze”: Concentrated on tongue exercises. Participants guide the character through a maze by using tongue movements. The character navigates accordingly by extending their tongue upwards, downwards, leftwards, or rightwards. The maze path corresponds to the intended tongue movement sequence, and the character exits the maze after successfully completing the routine. Each tongue movement is repeated 15 times. (3) “Leaping Barriers”: Emphasized Chin Tuck Against Resistance (CTAR) exercises. In this game, participants control the bird's flight by lowering their jaw. They press a rubber ball with their chin, hold for 2 or 3 s, and then

release, causing the bird to descend and maneuver around obstacles. Participants must navigate the bird around each obstacle a total of 15 times (see Fig. 1).

2.5.2. Usual care group

The control group received usual care treatment, including the following: (1) lip exercises: lip opening, lip closing, drumming cheeks, left drumming cheeks, and right drumming cheeks; (2) tongue exercises: sticking out the tongue, tongue going up, tongue going down, tongue going to the left, and tongue going to the right; (3) CTAR: using the chin to squeeze a rubber ball placed on the neck. The usual care treatment in the control group was the same as in the intervention group but without the FR-VG system.

2.6. Data collection and follow-up

After written consent was obtained from the participants, the assessor collected the baseline data from patients (T0), including sociodemographic information such as disease diagnosis, duration of disease, age, gender and medical history. Within 3 days of completing the 4-week intervention (T1), the outcome data were measured, and the AI-VG group was required to additionally complete the acceptance and satisfaction questionnaires. At 4 weeks after the end of the intervention (T2), the outcome data were measured for a third time to assess the short-term effect of the intervention.

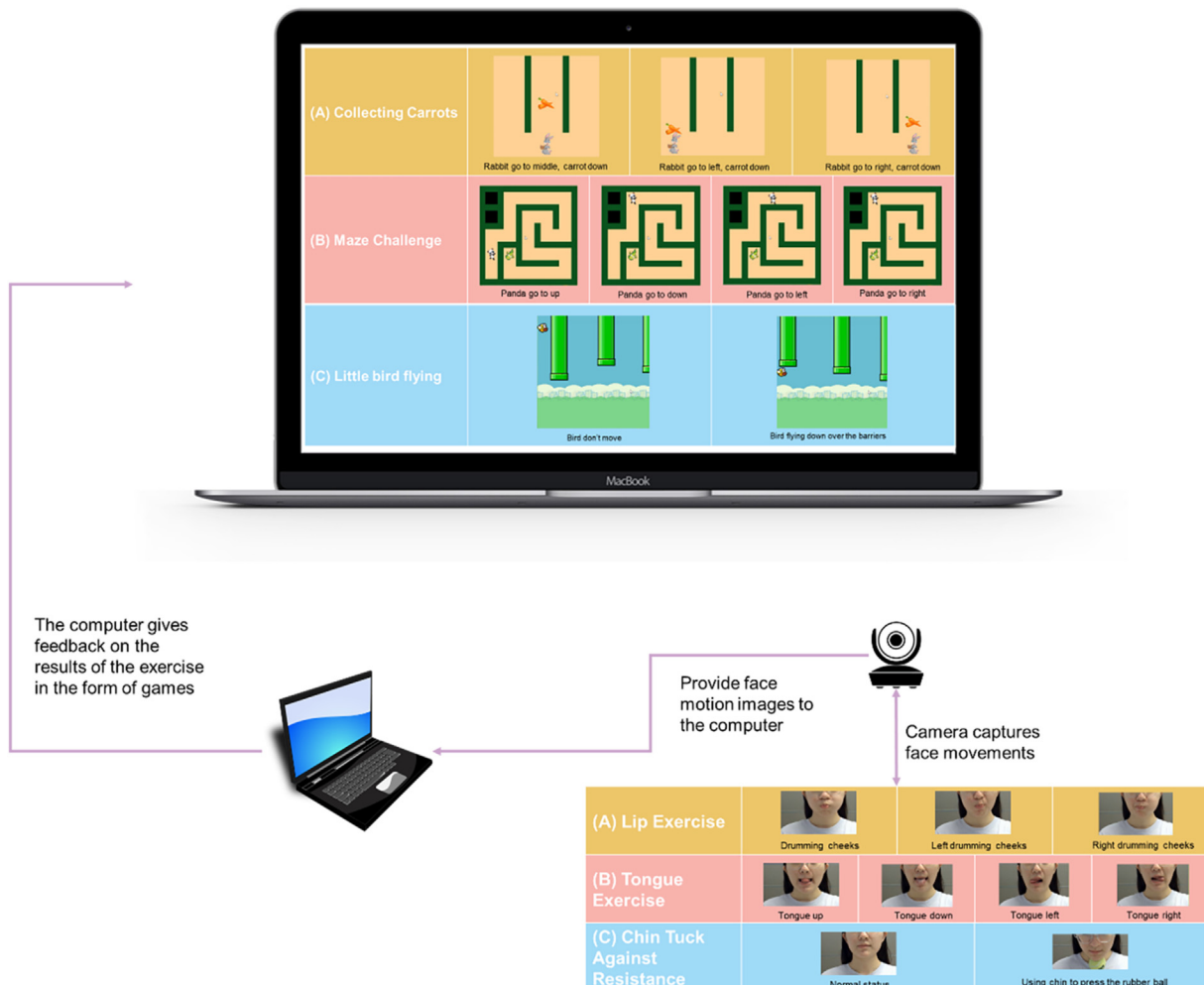


Fig. 1. Artificial Intelligence-based video games system game page.

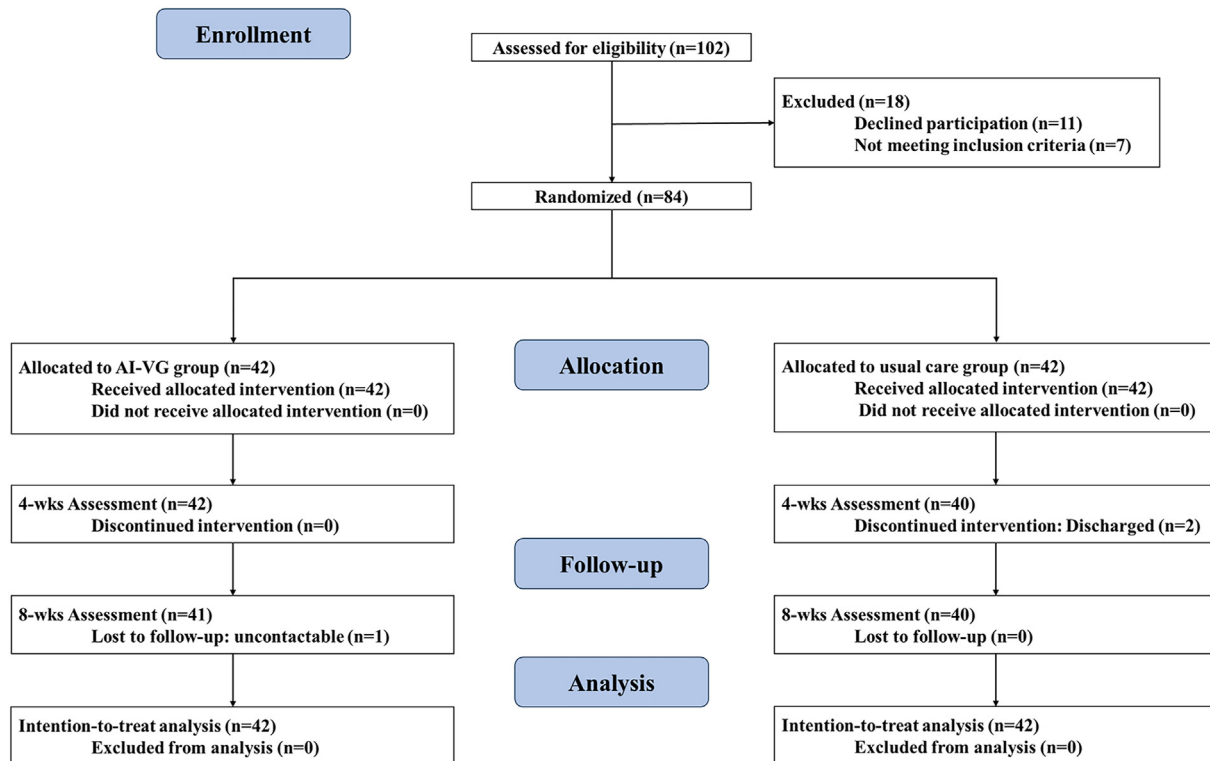


Fig. 2. CONSORT Flowchart.

2.7. Outcome measures

The primary outcome was swallowing function measured by GUSS. Swallowing function was assessed using water and other consistencies, with higher scores indicating better swallowing function in patients [24]. GUSS is the best test tool for dysphagia assessment after stroke, with a pooled sensitivity of 0.97 and a pooled specificity of 0.67 [25].

The secondary outcomes included the Standard Swallowing Assessment (SSA), which evaluates laryngeal function and pharyngeal function. The SSA scale scores ranged from 18 to 46, with higher scores indicating more serious subjective dysphagia [26]. The Functional Oral Intake Scale (FOIS) was used to assess the functional level of oral intake of food and liquid; it consisted of seven levels, with higher levels indicating better oral intake functionality [27]. Nutritional status was assessed using the Mini-Nutritional Assessment Short Form (MNA-SF), which consists of six questions, with higher scores suggesting better nutritional status [28]. Swallowing-related quality of life was assessed using the Swallowing Quality-of-Life Questionnaire (SWAL-QOL), which consists of 11 swallowing-related quality of life domains, with higher scores reflecting worse swallowing-related quality of life [29,30].

The AI-VG group's acceptance and satisfaction with the AI-VG system were evaluated using self-developed questionnaires on the basis of the Technology Acceptance Model (TAM) [31]. The acceptance questionnaire consists of seven dimensions, with scores ranging from 21 to 105. Higher scores indicated greater acceptance of the AI-VG system by the participants. The satisfaction questionnaire consisted of three dimensions, with a maximum score of 75. Higher scores reflected higher participant satisfaction with the AI-VG system.

The adherence to the intervention was quantified by the number of completed sessions out of the total 20 scheduled interventions.

Any adverse events that occurred while implementing the intervention were recorded and reported to ensure participant safety.

2.8. Statistical analysis

Statistical analyses were performed using IBM SPSS (version 29.0), with intention-to-treat analysis. Data normality was tested using normal Q–Q plots and Shapiro–Wilk tests. Normal continuous data were reported as the mean (SD) and analyzed using independent sample *t*-test. The median (interquartile range, IQR) and Mann–Whitney U tests were used for non-normal data. Categorical variables were compared using χ^2 test with frequencies and percentages. Generalized estimating equation (GEE) analysis was used to assess changes in the differences in primary and secondary outcomes at T1 and T2 compared with those in T0 between the two groups and compare changes in outcomes over time [32]. All statistical analyses were two-sided, and the significance level was set at 0.05.

3. Results

3.1. Characteristics of participants

As shown in Fig. 2, 84 participants met the inclusion criteria and provided a written informed consent. They were randomized to the AI-VG group ($n = 42$, 57.10 % male, age = 64.98 ± 9.66 years) and the usual care group ($n = 42$, 69.00 % male, age = 66.43 ± 13.12 years). One patient in the AI-VG group was dropped at T2 due to being uncontactable. Two patients in the usual care group were discharged, and they stopped the intervention at T1. At T0, no statistical differences were found between the AI-VG group and the usual care group in terms of demographic data, primary outcome, and secondary outcomes (Table 1).

3.2. Primary outcome

After the intervention, the mean GUSS scores in the two groups continued to increase from T0 to T1 and T2. Compared with the control group, the AI-VG group had significantly improved GUSS scores at T1 (mean difference = 4.02 ± 1.09 , 95 % CI = from -6.16 to -1.89 , $P < 0.001$; Table 2) and T2 (mean difference = 4.14 ± 1.03 , 95 % CI = from -6.16 to 2.12 , $P < 0.001$; Table 2 and eFig. 1 in Appendix A). The GEE modeling showed significant interaction terms (group $\times$ time) at T1 ($\beta = 4.38$, 95 % CI = 3.44 – 5.32 , $P < 0.001$) and T2 ($\beta = 5.05$, 95 % CI = 4.02 – 6.07 , $P < 0.001$; Table 3). This finding validated that AI-VG was effective in improving the swallowing function of patients with stroke.

3.3. Secondary outcomes

The AI-VG group had lower SSA scores at T1 (mean difference = -1.48 ± 0.90 , 95 % CI = from -0.28 to 3.24 , $P = 0.100$) and T2 (mean difference = -0.83 ± 0.86 , 95 % CI = from -0.84 to 2.51 , $P = 0.330$) than the usual care group (Table 2 and eFig. 2 in Appendix A), with no statistical difference. A significant group and time interaction was observed at T1 ($\beta = -1.76$, 95 % CI = from -3.02 to -0.50 , $P = 0.006$), but no statistically significant interaction effects were detected at T2 (Table 3).

The AI-VG group showed an additional increase in the FOIS scores of 1.07 (95 % CI = from -1.68 to 0.46) points at T1 and an increase of 1.19 (95 % CI = from -1.76 to -0.62) points at T2 (overall group $\times$ time interaction, $P < 0.001$; Table 2, Table 3, and eFig. 3 in Appendix A).

The AI-VG group had a significantly improved MNA-SF score at T1 (mean difference = 1.00 ± 0.47 , 95 % CI = from -1.92 to -0.08 , $P = 0.034$; Table 2 and eFig. 5 in Appendix A) compared with the usual care group. A significant interaction between group and time

was observed at T1 ($\beta = 0.69$, 95 % CI = 0.10 – 1.28 , $P = 0.023$), but no statistically significant interaction was found at T2 (Table 3).

The AI-VG group showed a more significant decrease in SWAL-QOL at T1 and T2 than the usual care group, with mean differences of -16.43 ± 4.05 (95 % CI = 8.50 – 24.36 , $P < 0.001$) and -21.00 ± 3.51 (95 % CI = 14.14 – 27.87 , $P < 0.001$; Table 2 and eFig. 4 in Appendix A). The GEE models for the interaction terms (group $\times$ time) for T1 ($\beta = -25.93$, 95 % CI = from -35.90 to -15.96 , $P < 0.001$) and T2 ($\beta = -30.50$, 95 % CI = from -40.26 to -20.74 , $P < 0.001$; Table 3) were significant.

3.4. Adherence, acceptance, satisfaction, and safety

The median (IQR) training adherence was significantly higher in the AI-VR group than in the usual care group (18.00 [17.00 – 20.00] vs. 16.00 [15.00 – 17.00], $P < 0.001$; Table 2). Only the AI-VG group completed the acceptance and satisfaction questionnaires, scoring 103.00 [100.50 – 104.00] and 73.00 [72.00 – 74.00], respectively (Table 2). One AI-VG group participant reported dizziness during participation in the trial, and such dizziness spontaneously subsided after stopping the intervention. The participant resumed participation the next day without re-reporting any discomfort such as dizziness.

4. Discussion

This RCT contributes to the limited research on the efficacy of AI-VG interventions among patients with post-stroke dysphagia. The findings provide compelling evidence for the effectiveness of AI-VG intervention in improving swallowing function and swallowing-related quality of life. Furthermore, the high levels of patient acceptance and satisfaction with the AI-VG system are noteworthy.

Table 1
Demographic data of the participants.

Variables	Total (n = 84, %)	AI-VG group (n = 42, %)	Conventional group (n = 42, %)	t/Z/ χ^2	P-Value
Diagnosis					
Ischemic Stroke	69 (82.10)	35 (83.30)	34 (81.00)	0.08	0.776
Hemorrhagic Stroke	15 (17.90)	7 (16.70)	8 (19.00)		
Disease duration (months)	Mean $\pm$ SD	1.72 $\pm$ 1.50	1.67 $\pm$ 1.48	-0.28	0.778
Gender				1.28	0.258
Male	53 (63.10)	24 (57.10)	29 (69.00)		
Female	31 (36.90)	18 (42.90)	13 (31.00)		
Age (years)	Mean $\pm$ SD	65.70 $\pm$ 11.48	64.98 $\pm$ 9.66	-0.58	0.565
Occupation type				3.26	0.515
Mainly mental work	9 (10.70)	5 (11.90)	4 (9.50)		
Mainly physical work	5 (6.00)	4 (9.50)	1 (2.40)		
Part mental, part physical	9 (10.70)	3 (7.10)	6 (14.30)		
Unemployed	5 (6.00)	2 (4.80)	3 (7.10)		
Retirement	56 (66.70)	28 (66.70)	28 (66.70)		
Marital status				4.96	0.175
Married	75 (89.30)	39 (92.90)	36 (85.70)		
Widowhood	3 (3.60)	0 (0.00)	3 (7.10)		
Divorced	3 (3.60)	2 (4.80)	1 (2.40)		
Unmarried	3 (3.60)	1 (2.40)	2 (4.80)		
Residence				0.81	0.369
Rural/suburban	52 (61.90)	28 (66.70)	24 (57.10)		
Towns/urban	32 (38.10)	14 (33.30)	18 (42.90)		
Smoking history	28 (33.30)	14 (33.30)	14 (33.30)	0.00	1.000
Drinking history	24 (28.60)	11 (26.20)	13 (31.00)	0.23	0.629
Hypertension	68 (81.00)	32 (76.20)	36 (85.70)	1.24	0.266
Diabetes	42 (50.00)	21 (50.00)	21 (50.00)	0.00	1.000
Coronary heart disease	16 (19.00)	7 (16.70)	9 (21.40)	0.31	0.578
Hyperlipidemia	30 (35.70)	15 (35.70)	15 (35.70)	0.00	1.000
Gastroesophageal reflux disease	16 (19.00)	8 (19.00)	8 (19.00)	0.00	1.000
Primary outcome	GUSS, median (IQR)	7.00 (3.00–11.00)	6.50 (3.00–11.25)	-0.10	0.921
Secondary outcomes	SSA score, median (IQR)	34.00 (29.25–36.00)	34.00 (30.00–35.25)	-0.20	0.840
	FOIS, median (IQR)	4.00 (2.00–4.00)	4.00 (2.00–5.00)	-0.65	0.520
	MNA-SF, median (IQR)	8.00 (6.00–10.00)	8.00 (6.00–10.00)	-0.57	0.570
	SWAL-QOL, mean $\pm$ SD	128.56 $\pm$ 23.62	133.31 $\pm$ 27.64	1.87	0.065

Abbreviation: AI-VG, artificial intelligence-based video-game; GUSS, Gugging Swallowing Screen; SSA, Standard Swallowing Assessment; FOIS, Functional Oral Intake Scale; MNA-SF, Mini-Nutritional Assessment Short Form; SWAL-QOL, Swallowing Quality-of-Life Questionnaire.

Table 2Outcome differences between both groups at T0 and at T1 and T2^a according to the general estimating equation.

Outcome by group		Baseline			Post-intervention			Follow-up		
		Mean ± SD	Difference between groups (95 % CI)	P	Mean ± SD	Difference between groups (95 % CI)	P	Mean ± SD	Difference between groups (95 % CI)	P
Primary outcome	GUSS score, mean ± SD									
	Intervention group	7.71 ± 0.78	0.19 ± 1.10 (−2.34, 1.96)	0.862	15.93 ± 0.73	4.02 ± 1.09 (−6.16, −1.89)	<0.001	16.71 ± 0.64	4.14 ± 1.03 (−6.16, −2.12)	<0.001
Secondary outcome	Control group	7.52 ± 0.77			11.90 ± 0.80			12.57 ± 0.81		
	SSA score, mean ± SD									
	Intervention group	32.71 ± 0.58	0.29 ± 0.85 (−1.94, 1.37)	0.735	25.43 ± 0.69	−1.48 ± 0.90 (−0.28, 3.24)	0.100	24.69 ± 0.66	−0.83 ± 0.86 (−0.84, 2.51)	0.330
	Control group	32.43 ± 0.62			26.90 ± 0.57			25.52 ± 0.55		
	FOIS score, mean ± SD									
	Intervention group	3.50 ± 0.20	0.17 ± 0.31 (−0.78, 0.45)	0.593	5.71 ± 0.23	1.07 ± 0.31 (−1.68, −0.46)	0.001	5.98 ± 0.20	1.19 ± 0.29 (−1.76, −0.62)	<0.001
	Control group	3.33 ± 0.24			4.64 ± 0.21			4.79 ± 0.21		
	MNA-SF score, mean ± SD									
	Intervention group	8.24 ± 0.34	0.31 ± 0.45 (−1.20, 0.58)	0.493	9.88 ± 0.34	1.00 ± 0.47 (−1.92, −0.08)	0.034	9.79 ± 0.34	0.79 ± 0.48 (−1.73, 0.16)	0.102
	Control group	7.93 ± 0.30			8.88 ± 0.32			9.00 ± 0.34		
	SWAL-QOL score, mean ± SD									
	Intervention group	133.31 ± 4.21	9.50 ± 5.02 (−19.33, 0.33)	0.058	81.74 ± 3.22	−16.43 ± 4.05 (8.50, 24.36)	<0.001	71.90 ± 2.57	−21.00 ± 3.51 (14.14, 27.87)	<0.001
	Control group	123.81 ± 2.72			98.17 ± 2.45			92.90 ± 2.38		
Adherence, day, median (IQR)	Intervention group	NA	NA	NA	18.00 (17.00–20.00)	5.452 ^b	<0.001	NA	NA	NA
	Control group	NA	NA	NA	16.00 (15.00–17.00)			NA	NA	NA
Acceptability score, median (IQR)	Intervention group	NA	NA	NA	103.00 (100.50–104.00)	NA	NA	NA	NA	NA
	Control group	NA	NA	NA				NA	NA	NA
Satisfied score, median (IQR)	Intervention group	NA	NA	NA	73.00 (72.00–74.00)	NA	NA	NA	NA	NA
	Control group	NA	NA	NA				NA	NA	NA

Abbreviation: GUSS, Gugging Swallowing Screen; SSA, Standard Swallowing Assessment; FOIS, Functional Oral Intake Scale; MNA-SF, Mini-Nutritional Assessment Short Form; SWAL-QOL, Swallowing Quality-of-Life Questionnaire; NA, not applicable.

^a T0 is before the intervention. T1 is after the intervention. T2 is one month follow up.

^b Used the Mann–Whitney U test rather than general estimating equation.

Table 3Changes in primary and secondary outcomes at T1 and T2 in two groups compared to T0^a based on GEE models.

	β	95%CI		<i>P</i>
		Lower limits	Upper limits	
GUSS				
Group (AI-VG group vs usual care group)	0.19	−1.96	2.34	0.862
Time				
Baseline	Reference			
4th W	4.38	3.44	5.32	<0.001
8th W	5.05	4.02	6.07	<0.001
Group * time				
Group * T0	Reference			
Group * T1	3.83	2.20	5.47	<0.001
Group * T2	3.95	2.30	5.61	<0.001
SSA				
Group (AI-VG group vs usual care group)	32.43	32.22	33.64	<0.001
Time				
Baseline	Reference			
4th W	−5.52	−6.14	−4.91	<0.001
8th W	−6.91	−7.73	−6.08	<0.001
Group * time				
Group * T0	Reference			
Group * T1	−1.76	−3.02	−0.50	0.006
Group * T2	−1.12	−2.45	0.22	0.100
FOIS				
Group (AI-VG group vs usual care group)	3.33	2.87	3.80	<0.001
Time				
Baseline	Reference			
4th W	1.31	1.08	1.54	<0.001
8th W	1.45	1.23	1.67	<0.001
Group * time				
Group * T0	Reference			
Group * T1	0.91	0.57	1.24	<0.001
Group * T2	1.02	0.68	1.36	<0.001
MNA-SF				
Group (AI-VG group vs usual care group)	7.93	7.34	8.51	<0.001
Time				
Baseline	Reference			
4th W	0.95	0.51	1.40	<0.001
8th W	1.07	0.62	1.52	<0.001
Group * time				
Group * T0	Reference			
Group * T1	0.69	0.10	1.28	0.023
Group * T2	0.48	−0.13	1.08	0.122
SWAL-QOL				
Group (AI-VG group vs usual care group)	123.81	118.47	129.15	<0.001
Time				
Baseline	Reference			
4th W	−25.64	−32.26	−19.02	<0.001
8th W	−30.91	−37.27	−24.54	<0.001
Group * time				
Group * T0	Reference			
Group * T1	−25.93	−35.90	−15.960	<0.001
Group * T2	−30.50	−40.26	−20.740	<0.001

Abbreviation: AI-VG, artificial intelligence-based video-game; GUSS, Gugging Swallowing Screen; SSA, Standard Swallowing Assessment; FOIS, Functional Oral Intake Scale; MNA-SF, Mini-Nutritional Assessment Short Form; SWAL-QOL, Swallowing Quality-of-Life Questionnaire; GEE, general estimating equation.

^a T0 is before the intervention. T1 is after the intervention. T2 is one month follow up.

4.1. Improvement in swallowing function

Intervention with AI-VG significantly improved swallowing function and increased oral intake level in patients with post-stroke dysphagia, consistent with previous studies [18,33]. The AI-VG system allows participants to visualize their actions through the computer screen during training, boosting self-confidence and aiding swallowing function recovery [34]. The interaction of AI and video games can improve facial action recognition accuracy and provide rapid real-time feedback [35].

However, no significant difference in SSA score changes was found between the AI-VG and usual care groups. This discrepancy may be attributed to SSA's focus on assessing laryngeal reflex ability with water, primarily evaluating aspiration risk rather than

comprehensive dysphagia symptoms. By contrast, GUSS and FOIS assessments encompass multiple food types and guide safe food reintroduction [36]. These findings underscore the need for further exploration of AI-VG-based dysphagia therapies.

4.2. Impact on nutrition and quality of life

The results showed that the AI-VG group had significantly improved nutritional levels and swallowing-related quality of life compared with the usual care group, aligning with findings from Battel et al. [37]. The enhancement of swallowing function through AI-VG likely diversifies patients' dietary choices and improves eating experiences, leading to psychological fulfillment and improvements in swallowing-related quality of life and nutritional

status. Additionally, nutritional status may indirectly affect quality of life. Improved nutritional status could prevent complications, such as weight loss, limited mobility, and fatigue, thus increasing social participation and consequently improving quality of life [38].

However, the analysis revealed a significant between-group difference in MNA scores at T1, which subsequently normalized at T2. This temporal pattern suggests a potential early acceleration effect of the AI-VG intervention on nutritional status improvement. The initial advantage observed in the AI-VG group may reflect enhanced early engagement in swallowing exercises, leading to more rapid improvement in oral intake capacity. However, the convergence of scores at T2 indicates that both interventions ultimately achieve comparable long-term nutritional outcomes, consistent with previous studies demonstrating similar plateauing effects in dysphagia rehabilitation [39,40]. This finding has important clinical implications, suggesting that AI-VG may be particularly valuable during the critical early recovery period when accelerated improvement could reduce complications associated with prolonged dysphagia.

4.3. Adherence, satisfaction, and acceptance

The intervention group had significantly higher training adherence than the control group, and their satisfaction and acceptance of the AI-VG intervention was high. This finding aligns with Park et al.'s findings [17], which reported higher motivation, interest, and lower physical and muscle fatigue in the game group than in the control group. The accessibility of the proposed AI-VG system, requiring only a computer with a camera, may facilitate patient immersion in the training process with reduced burden, potentially contributing to enhanced acceptance and satisfaction.

4.4. Implications for research and clinical practice

Rehabilitating dysphagia is a complex process that demands a multidisciplinary team effort, including speech-language pathologists, therapists, doctors, and nurses [9,41]. Effective collaboration with patients and their families, focusing on their values and needs, is essential for successful treatment. Evidence suggests that the use of game-based training can enhance patient engagement and motivation [42]. Real-time feedback on training results can be provided by integrating games with computers and artificial intelligence, thereby improving the efficiency of rehabilitation. The present study demonstrated that the use of AI-VG to assist patients with swallowing rehabilitation could be effective and feasible, and it could fill the gap in the use of intelligent methods for swallowing rehabilitation in patients with post-stroke dysphagia.

In the future, AI-VG could incorporate near-infrared spectroscopy (NIRS), thus providing complementary insights into the peripheral and central mechanisms involved in post-stroke dysphagia rehabilitation. While AI-based facial recognition methods can effectively capture external muscle movements, NIRS can directly measure muscle activation patterns, and monitor prefrontal cortex activity during swallowing tasks [43]. NIRS measurements of muscle activation can validate and complement the facial recognition results in this study, thus providing quantitative data on muscle recruitment intensity and timing, further understanding which specific muscle patterns are most indicative of successful rehabilitation outcomes, and providing patients with more personalized approach to treatment. Future studies should incorporate standardized and comprehensive baseline stroke assessments, such as the National Institutes of Health Stroke Scale (NIHSS) or detailed neuroanatomical characterization of lesion location and extent, to ensure enhanced matching between intervention and control groups and analyze the potential impact of

stroke severity or stroke regions on treatment outcomes. Understanding the relationship between cortical activation patterns and successful rehabilitation outcomes could help identify optimal intervention strategies for different stroke subtypes and severity levels.

4.5. Strengths and limitations

This study has several strengths. First, it utilized an RCT design, widely recognized as the gold standard for evaluating intervention efficacy. Furthermore, the single-blind scoring process significantly mitigates potential bias, thereby enhancing the reliability of the findings. Second, to the authors' knowledge, this study is the first to use an AI-based video game for rehabilitation in patients with post-stroke dysphagia. This innovative approach not only addresses a critical gap in the literature but also provides valuable insights into the development of telemedicine and community-based rehabilitation strategies. The AI-VG system's potential for remote application could prove particularly beneficial in situations where in-person therapy is challenging or impossible.

This study also has some limitations to consider. First, this study was conducted in one rehabilitation hospital, which may limit the generalizability of the findings. A multicenter approach with a larger sample size could have provided more robust and widely applicable results. Future research should consider expanding to diverse healthcare settings and geographical locations to enhance external validity. Second, the technological experience or the educational levels of participants were not assessed before the intervention. This oversight may have introduced bias in the evaluation of adherence, acceptance, and satisfaction with the AI-VG system. Participants with greater technological proficiency may have found the system more acceptable, and patients with different educational levels may experience varying degrees of comfort and competency with technological interfaces, which could affect their engagement with and adherence to the intervention, potentially skewing the results. Future studies should incorporate baseline assessments of educational background and technological literacy to account for these potential confounding factors. This incorporation could enable more precise analysis of the relationship between these variables and treatment outcomes, ultimately informing the development of more accessible and effective rehabilitation protocols. Third, blinding participants and implementers was not possible due to the nature of the AI-VG system. This awareness could have potentially influenced participants' performance and implementers' behavior. While challenging to overcome in studies involving obvious technological interventions, future research may consider using active control groups with sham interventions to mitigate this limitation. Forth, this study focused primarily on short-term outcomes, thus limiting the understanding on the long-term efficacy of the AI-VG intervention. Future research should incorporate extended follow-up periods (e.g., 6 months or 1 year) to assess the long-term impact and sustainability of the AI-VG intervention. Fifth, stroke severity at baseline was not assessed using standardized measures such as NIHSS. Moreover, the affected brain region and areas after stroke, which could potentially influence dysphagia recovery outcomes, were not assessed. Future studies should include assessments of stroke severity and affected brain regions and areas to better stratify participants and identify which patients may benefit most from AI-VG intervention.

5. Conclusions

This study presents promising evidence for the effectiveness of AI-VG interventions in improving swallowing function, quality of life, and nutritional status among patients with post-stroke

dysphagia. The high level of patient satisfaction further supports the potential of this approach. However, larger and more diverse studies are needed to conclusively establish the role of AI-VG in dysphagia rehabilitation and guide its implementation in clinical settings. As technology continues to advance, AI-VG interventions may represent an important frontier in the evolving landscape of stroke rehabilitation.

CRediT authorship contribution statement

Bohan Zhang: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing-Original Draft, Visualization. Ka Po Wong: Conceptualization, Methodology, Writing-Review&Editing. Mingyue Liu: Methodology, Investigation, Resources, Validation. Vivian Hui: Methodology, Writing-Review&Editing. Cai Guo: Software, Data Curation. Yue Liu: Conceptualization, Methodology, Validation, Investigation, Resources. Zihan Liu: Investigation, Resources. Qian Xiao: Validation, Methodology, Resources, Writing-Review&Editing, Funding acquisition. Jing Qin: Methodology, Resources, Writing-Review&Editing, Supervision.

Data availability statement

The datasets generated and analyzed during the current study are not publicly available due to privacy protection and ethical considerations, but are available from the corresponding author on reasonable request.

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Conflict of interest

The authors declare no potential conflicts of interest regarding the authorship or publication of this article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clnu.2024.12.022>.

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